Single Cycle Check

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The problem

- Each element is a jump (wrapping).
- Do the jumps form one cycle visiting every index exactly once and returning to the start?

Intuition

- Follow the jumps n times: we must visit every index exactly once and return to the start.

Approach

- 1. Jump n times with wrap-around
- 2. If we return to zero early, fail
- 3. After n jumps we must be at zero

Java solution

```java
public class SingleCycleCheck {
    public static boolean hasSingleCycle(int[] a) {
        int n = a.length, idx = 0;
        for (int visited = 0; visited < n; visited++) {
            if (visited > 0 && idx == 0) return false;
            idx = Math.floorMod(idx + a[idx], n);
        }
        return idx == 0;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Follow the jumps n times: we must visit every index exactly once and return to the start.
- Complexity: O(n)